Hadi Khalaf

hadikhalaf@g.harvard.edu hadikhalaf.com

Education

Harvard University 2024 – Expected 2028
Ph.D. in Computer Science

- Designing reliable and safe AI systems under weak supervision
- Supported by the Harvard Graduate Prize Fellowship and the Kempner Institute Graduate Fellowship
- *Committee:* Flavio Calmon *(primary advisor)*, Boaz Barak, and Ariel Procaccia

Harvard University 2024 – 2026
SM in Computer Science, GPA: 3.97/4.0

- *Selected coursework:* reinforcement learning, high-dimensional statistics, nonasymptotic statistics (MIT), optimization (MIT)

American University of Beirut 2020 – 2024
BS in Statistics, BE in Electrical & Computer Engineering with Highest Distinction, GPA: 4.19/4.0

- Graduated as valedictorian of the Faculty of Engineering and Architecture based on merit and character
- Supported by the AUB Presidential Merit Scholarship (one of the top six undergraduate applicants across the university)

Publications and Preprints *(* indicates first authorship with equal contribution)*

Circuits of LLM Classifiers *Under review*
HK*, Sajani Vithana*, Kunal Talreja, Haewon Jeong, Viveck Cadambe, Flavio Calmon

Robust AI Evaluation through Maximal Lotteries ICML 2026
HK, Serena Wang, Daniel Halpern, Itai Shapira, Flavio Calmon, Ariel Procaccia

Inference-Time Reward Hacking in Large Language Models NeurIPS 2025 (Spotlight)
HK, Claudio Mayrink Verdun, Alex Oesterling, Hima Lakkaraju, Flavio Calmon

AI Alignment at Your Discretion ACM FAccT 2025, Best Paper at NENLP
Maarten Buyl*, **HK***, Claudio Mayrink Verdun*, Lucas Monteiro Paes*, Caio Vieira Machado, Flavio Calmon

Experience

Microsoft June 2026 – August 2026
Applied Scientist Intern

- Designed an end-to-end pipeline for autonomously building and optimizing agentic skills from task specifications
- Improved research-grade math and Excel manipulation performance by 38% with training cost under $5

Beirut Urban Lab Sep 2023 – May 2024
Machine Learning Engineer

- Built and automated an end-to-end agentic pipeline for data elicitation, storage, and retrieval
- Deployed the first GPT-powered tool to crowdsource and query geospatial data over WhatsApp and ArcGIS
- Boosted user engagement with surveys by 72% and user retention by 48%

Harvard University Jun 2023 – Sep 2023
Research Fellow, Economics Department

- Helped develop the first Bayesian partial identification algorithms for two-player games
- Implemented game-theoretic algorithms to solve two-player games in underspecified economic setups

Massachusetts Institute of Technology Jun 2019 – August 2019
Research Science Institute Fellow, Nuclear Engineering Department

- Modeled a micro-nuclear reactor on CASMO 4e with a cost per kWh competitive to fossil fuels
- Employed a novel 3D-printed fuel to optimize yield rates and thermodynamic specifications

Teaching

Harvard ES 250: Information Theory Fall 2026
AUB EECE 230: Introduction to Programming (Course manager and Head TA) Fall 2022 & Spring 2022, 2023, 2024
AUB EECE 330: Data Structures and Algorithms (Course manager and Head TA) Fall 2023
AUB EECE 210: Electric Circuits (Head TA) Fall 2021

Awards and Fellowships

Kempner Institute Graduate Fellowship 2026
Awarded to select graduate students across Harvard working on natural and artificial intelligence

ICML 2026 Silver Reviewer Award 2026

Harvard Berkman Klein Student Leader in AI 2026

NeurIPS 2025 Scholar Award 2025

Princeton Machine Learning Theory Summer School Fellowship 2025

ICML 2025 Hudson River Trading Travel Award 2025

Best Paper Award at the New England NLP Workshop at Yale 2025

Harvard AI Safety Team Technical Fellowship 2024

Harvard Graduate Prize Fellowship 2024
Awarded to select incoming students across Harvard Graduate School of Arts and Sciences based on intellectual promise

AUB High Distinction Award 2024

AUB Penrose Award in the Faculty of Engineering and Architecture 2024
Valedictorian, awarded to the most outstanding graduating student on the basis of scholarship and character

AUB Presidential Scholarship 2020
Full four-year scholarship awarded to the six most outstanding applicants across AUB (0.4% acceptance)

Research Science Institute Fellow at MIT 2019
Hosted as a researcher in the Nuclear Science and Engineering Department at MIT (2% acceptance)

Nationwide Winner in the Lebanese Physics Olympiad 2019
Ranked first among all 11th graders in Lebanon; unable to compete at IPhO since Lebanon is not represented there

Invited Talks

Stanford University AI Evaluation Seminar November 2026

Max Planck Institute for Software Systems October 2026

Harvard CS 2901 PhD Seminar September 2026

Harvard CS 2902 PhD Seminar March 2026

New England NLP Workshop at Yale March 2025